

MINOR PROJECT

Social Media Usage Analysis for Student Stress Detection and Management System

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Course

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Project Guide

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Problem Scoping

Social media has become an essential part of students' daily lives. While it provides communication, entertainment, and learning opportunities, excessive usage can negatively impact mental health, sleep quality, academic performance, and overall well-being. Educational institutions often face challenges in identifying students who may be experiencing stress due to unhealthy social media habits.

The problem addressed in this project is to analyze social media usage patterns and predict student stress levels using data-driven techniques. By examining factors such as daily usage hours, late-night activity, screen time before sleep, mood score, and academic performance, the system helps identify students who may require additional support and stress management interventions.

Dataset Creation

Since real student stress data is difficult to obtain due to privacy concerns, a synthetic dataset of 10,000 student records was generated using Python. The dataset simulates realistic student behavior related to social media usage, sleep patterns, academic performance, and emotional well-being.

Dataset Attributes

- Student ID

- Age
 - Gender
 - Daily Usage Hours
 - Late Night Usage
 - Number of Platforms Used
 - Posts Per Week
 - Screen Time Before Sleep
 - Sleep Hours
 - Academic Performance
 - Mood Score
 - Risk Score
 - Stress Level
-

Project Objective

The objective of this project is to develop a Social Media Usage and Stress Management System that analyzes student social media behavior and predicts stress levels. The project aims to create a structured dataset, identify key stress-related factors, and generate meaningful insights that can assist counselors and educational institutions in promoting student mental well-being and effective stress management.

Use Case: Predicting Student Stress Levels Based on Social Media Usage

A large university has observed increasing social media usage among students and growing concerns regarding stress, anxiety, sleep disturbances, and academic performance. By analyzing students' social

media habits and related behavioral factors, the university aims to identify stress patterns and develop strategies to improve student well-being and academic success.

Problem Statement

The university administration wants to improve student mental health monitoring. They approach an AI team and ask:

'Build an AI system that can automatically analyze social media usage patterns and identify students with high stress levels.'

Although the request sounds straightforward, the solution statement is not a well-defined AI problem. The challenge lies in converting the broad objective into a structured machine learning task by identifying relevant features, defining stress indicators, selecting suitable algorithms, and establishing measurable evaluation criteria.

Step 1: Understanding the Problem

Initial Statement

Student stress is increasing, but the impact of social media usage on stress levels is not clearly understood.

Questions for University Administration

- How many students are enrolled in the university?
- How frequently are student stress assessments conducted?
- What is the current process for identifying stressed students?
- Which stress-related issues create the biggest problems when not identified early?

Questions for Counsellors

- How do you currently identify students experiencing stress?
- Which student cases consume the most counseling time?
- What challenges do you face while reviewing student wellness reports?
- Are there delays in identifying students who need immediate support?

Questions for Students

- How many hours do you spend on social media daily?
- Have you experienced stress due to excessive social media usage?
- Which social media activities affect your mental well-being the most?
- Are you satisfied with your current social media habits?
- What is the biggest issue you face because of social media usage?

Key Findings from Initial Investigation

- Around **10,000 students** are enrolled in the university.
- Stress assessments are conducted **once per semester**.
- Average social media usage is approximately **5–6 hours per day**.
- The desired goal is **early identification of high-stress students**.
- Approximately **30–35% of students** report moderate to high stress levels.

Insights from Student Welfare Department

- Counselors can review approximately **20–30 student cases per week**.
- Manual stress assessment consumes a significant amount of time and effort.
- The large student population makes continuous monitoring difficult.

These challenges highlight the need for an AI-based system that can automatically identify students at risk of high stress, enabling counselors to focus on students who require immediate attention.

Feedback from Counseling Department

- Student stress levels are assessed manually.
- Counselors can review only a limited number of students each week.

- High student volume causes delays in identifying at-risk students.
- Highly stressed students are sometimes overlooked.
- Different counselors may interpret stress levels differently, leading to inconsistent assessments.

Key Challenges Identified

- Time-consuming manual evaluation process.
- Limited counseling capacity.
- Delayed identification of high-risk students.
- Risk of missing students who require immediate support.
- Lack of consistency in stress assessment decisions.

Some more clarity about the problem statement

Step 2: Define the Problem Precisely

Poor Problem Statement

"Analyze social media usage."

Better Problem Statement

"Identify students with high stress levels based on their social media usage patterns so that counselors can provide timely support before stress leads to serious academic, emotional, or mental health consequences."

Who?

- Students using social media platforms
- Counselors / Student Welfare Department
- University Administration

What?

- Predict student stress level
- Classify stress levels as **High Stress**, **Medium Stress**, or **Low Stress**

When?

- After collecting social media usage and student wellness data
- During regular student assessments

Why?

- Enable early intervention and counseling support
- Improve student mental well-being
- Reduce stress-related academic and personal issues
- Allocate counseling resources more effectively

Step 3: Is AI Really Needed?

Simple Rule-Based Approach

- If social media usage > 8 hours/day → **High Stress**
- If social media usage < 2 hours/day → **Low Stress**
- If sleep duration < 5 hours/day → **Medium Stress**

So the Question Is: Why AI?

AI should only be used when traditional rule-based systems are not sufficient to solve the problem effectively.

AI is Needed When:

- Patterns are complex and difficult to define using simple rules.
- Multiple factors interact with each other.
- Manual rules cannot accurately identify all cases.

Factors Influencing Student Stress Levels

- Daily social media usage time
- Number of social media platforms used
- Sleep duration
- Screen time before bedtime
- Academic workload
- Online interactions and engagement patterns
- Lifestyle and wellness habits

Conclusion

Since student stress depends on multiple interconnected factors, a simple rule-based system may not be able to identify stress levels accurately. Therefore, an AI-based predictive model is more suitable for detecting high-risk students and enabling timely intervention.

Additional Factors Influencing Student Stress Levels

- Academic workload
- Online interactions and engagement
- Social comparison behavior
- Cyberbullying experiences
- Emotional sentiment of posts and comments
- Previous stress or mental wellness history
- Age and demographic factors
- Frequency of social media checking

These factors interact in complex ways and cannot be accurately modeled using simple if-else rules.

Machine Learning algorithms can analyze these multiple variables simultaneously, discover hidden patterns, and predict student stress levels more effectively.

Step 4: Identification of Stakeholders

Stakeholders are individuals or groups who are directly or indirectly affected by the project and its outcomes.

Stakeholder Mapping

- **University Administration** – Improve student well-being and support informed decision-making regarding student mental health.
- **Students / Users** – Receive timely counseling support and early stress intervention.
- **Counselors / Student Welfare Staff** – Identify students with high stress levels who require immediate attention.
- **Mental Health Professionals** – Provide appropriate guidance and intervention for at-risk students.
- **Parents / Guardians** – Gain awareness of factors affecting student well-being and support students when necessary.
- **Department Heads** – Monitor stress trends among students and improve academic support systems.
- **IT Department** – Integrate, maintain, and monitor the AI-based stress prediction system.
- **University** – Improve student satisfaction, academic performance, and overall mental wellness.

Step 5: Define the AI Task

Machine Learning Formulation

The objective is to predict the stress level of students based on social media usage patterns and wellness-related factors.

1. Classification Problem

Output Categories:

- High Stress

- Medium Stress
- Low Stress

Since the model predicts one of three possible classes, this is a **Multi-Class Classification Problem**.

2. Risk Scoring

The AI system can also estimate the probability or risk score of stress on a scale from **0% to 100%**.

Examples:

- Student A → 95% Stress Risk
- Student B → 65% Stress Risk
- Student C → 20% Stress Risk

3. Ranking Problem

After calculating stress risk scores, students can be ranked from highest risk to lowest risk. This helps counselors prioritize and determine which students should be contacted first for support and intervention.

AI Task Summary

- **Classification** → Predict stress category.
- **Risk Scoring** → Estimate stress probability.
- **Ranking** → Prioritize students for counseling support.

Step 6: Defining Success Metrics

Defining success metrics is critical because it helps evaluate how well the AI model predicts student stress levels and supports decision-making.

1. Accuracy

Measures how many student stress levels are correctly classified by the model.

2. Precision

Out of all students predicted as **High Stress**, how many were actually High Stress.

3. Recall

Out of all actual **High Stress** students, how many were successfully identified by the model.

4. F1 Score

Represents the balance between **Precision** and **Recall**. It is useful when both false positives and false negatives are important.

5. ROC-AUC

Measures the model's ability to distinguish between high-stress and low-stress students across different classification thresholds.

Success Criteria

- High classification accuracy.
- Reliable identification of high-stress students.

- Low number of missed at-risk students.
- Strong balance between Precision and Recall.
- Effective prioritization for counseling intervention.

Step 7: Defining Constraints

Every AI project operates under certain constraints that can affect the design, implementation, and performance of the system.

1. Operational Constraints

- Counselors can review only a limited number of students per week.
- Limited mental health professionals are available for student support.
- Counseling resources must be allocated efficiently to high-stress students.

2. Time Constraints

- Students at risk should be identified as early as possible.
- High-stress students must be detected before stress leads to serious academic or mental health consequences.
- The system should provide predictions within a few seconds.

3. Data Constraints

Certain information may be unavailable, incomplete, or inaccurate.

- Incomplete survey responses.
- Missing social media usage data.
- Lack of historical stress assessment records.
- Incorrect or inconsistent stress labels.
- Limited information about students' emotional well-being.

4. Ethical Constraints

Students should not be negatively affected by incorrect predictions. The AI system must be fair, reliable, and responsible.

- Can a highly stressed student be wrongly classified as Low Stress?
- How can student privacy and data security be protected?
- How can bias in predictions be minimized?
- How can transparency and fairness be ensured?
 - Can a highly stressed student be wrongly classified as Low Stress?
 - Can students request manual counseling assessment or review?
 - Is the model biased toward certain groups of students?
 - Are student privacy and personal data protected?
 - Can the system explain why a student received a particular stress level prediction?
 - Is sensitive student information kept confidential?
 - How can bias in predictions be minimized?
 - How can transparency and fairness be ensured?

Conclusion

Understanding operational, time, data, and ethical constraints helps build an AI system that is accurate,

Step 8 :- Assumptions & Risks

Assumptions

- Historical student behavior reflects future behavior patterns.
- Previously assigned stress labels are accurate.
- Social media usage data contains sufficient information for analysis.
- Counselors will act on the system's recommendations.
- Students provide genuine and accurate responses.

Risks

- Incomplete survey responses.
- Missing social media usage information.
- Incorrect historical stress labels.
- Changes in student social media behavior over time.
- High-stress students may be incorrectly classified as Low Stress.
- Bias in historical student data affecting model predictions.
- Sudden events causing stress patterns to change unexpectedly.

Importing Libraries

- **Pandas** - For data manipulation and analysis.
- **NumPy** - For numerical computations.

```
In [1]: import numpy as np # num - numerical py- python
import pandas as pd # as - alias
```

Generating Synthetic Student Social Media Usage Dataset

```
In [2]: np.random.seed(45)
n_student = 10000
data = {
    "student_id": [f"STU{i:05d}" for i in range(1, n_student+1)],
    "age": np.random.randint(18, 26, n_student),
    "gender": np.random.choice(["Male", "Female"], n_student),
    "daily_usage_hours": np.round(np.random.uniform(0.5, 10, n_student), 1),
    "late_night_usage": np.random.choice(["Yes", "No"], n_student),
    "number_of_platforms": np.random.randint(1, 7, n_student),
    "posts_per_week": np.random.randint(0, 30, n_student),
    "screen_time_before_sleep": np.random.randint(0, 180, n_student),
    "sleep_hours": np.round(np.random.uniform(4, 10, n_student), 1),
    "academic_performance": np.round(np.random.uniform(50, 95, n_student), 1),
    "mood_score": np.random.randint(1, 11, n_student)
}
df = pd.DataFrame(data)
```

```
In [3]: df
```

Out[3]:

	student_id	age	gender	daily_usage_hours	late_night_usage	number_of_platforms	posts_per_week	screen_time_before_sl
0	STU00001	21	Female	8.1	No	6	29	
1	STU00002	24	Male	0.5	Yes	1	27	
2	STU00003	21	Female	4.4	No	3	16	
3	STU00004	22	Male	3.7	Yes	3	22	
4	STU00005	18	Male	8.3	No	4	19	
...	...	...	...	...	...	...	...	...
9995	STU09996	19	Female	2.3	Yes	2	15	
9996	STU09997	20	Female	7.5	Yes	3	25	
9997	STU09998	18	Female	2.5	Yes	2	24	
9998	STU09999	22	Female	6.7	No	2	25	
9999	STU10000	24	Female	2.7	Yes	1	14	

10000 rows × 11 columns

Calculate Risk Score

```
In [4]: risk_score = (
    0.30 * (df["daily_usage_hours"] > 7).astype(int) +
    0.20 * (df["late_night_usage"] == "Yes").astype(int) +
    0.15 * (df["sleep_hours"] < 6).astype(int) +
    0.15 * (df["screen_time_before_sleep"] > 120).astype(int) +
    0.10 * (df["number_of_platforms"] > 4).astype(int) +
    0.10 * (df["mood_score"] < 5).astype(int)
)
#Adding random variation
risk_score += np.random.normal(0, 0.05, n_student)
```

```
df["risk_score"] = np.clip(risk_score * 100, 0, 100).astype(int)

# Stress Level
df["stress_level"] = np.where(
    df["risk_score"] >= 70, "High",
    np.where(df["risk_score"] >= 40, "Medium", "Low")
)

df.head()
```

```
Out[4]:
```

	student_id	age	gender	daily_usage_hours	late_night_usage	number_of_platforms	posts_per_week	screen_time_before_sleep
0	STU00001	21	Female	8.1	No	6	29	22
1	STU00002	24	Male	0.5	Yes	1	27	119
2	STU00003	21	Female	4.4	No	3	16	99
3	STU00004	22	Male	3.7	Yes	3	22	116
4	STU00005	18	Male	8.3	No	4	19	119

Export Dataset to CSV File

```
In [5]: df.to_csv("student_dropout_synthetic_data", index= False)
```